

ERP Integration & Azure Cloud Migration

Client: Major Regional Logistics Operator — Confidential

Supplier: NileTech Solutions

Document Classification: Historical Technical Proposal — Won

Proposal Reference: NTS-LGS-AZ-067/2023

Delivery Model: Hybrid Agile / Cloud Transformation Programme

Target Cloud: Microsoft Azure

Primary ERP: SAP S/4HANA

Programme Duration: 44 Weeks

Migration Strategy: Hybrid — Replatform / Refactor / Retire

1. Executive Summary & Project Scope

1.1 Executive Summary

NileTech Solutions was awarded a multi-phase **ERP integration and cloud migration programme** for a regional logistics operator managing freight forwarding, warehousing, fleet operations, customs clearance, invoicing and last-mile delivery.

The existing environment consisted of a centrally hosted ERP platform surrounded by multiple point-to-point integrations with:

- Warehouse Management Systems (**WMS**)
- Transport Management Systems (**TMS**)
- Fleet Telematics
- Customer portals
- EDI gateways
- Customs platforms
- Banking/payment interfaces
- Human Resources systems
- Legacy reporting databases

The existing architecture had evolved incrementally over several years, resulting in **integration duplication, inconsistent master data, limited observability and high operational dependency on legacy infrastructure**.

NileTech's proposed programme introduced an Azure-based integration and application platform while maintaining controlled connectivity to the existing ERP environment during transition.

1.2 Programme Objectives

The programme objectives were defined as follows:

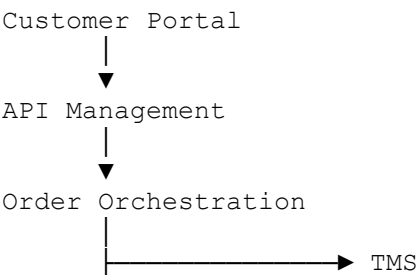
- 1. Migrate approved workloads from the existing data centre to **Microsoft Azure**.
- 2. Establish a centralized enterprise integration layer.
- 3. Reduce point-to-point ERP integrations.
- 4. Implement standardized API and event-driven integration patterns.
- 5. Establish centralized monitoring and operational telemetry.
- 6. Improve disaster recovery capabilities.
- 7. Implement Infrastructure-as-Code (**IaC**).
- 8. Establish automated CI/CD pipelines.
- 9. Introduce centralized identity and secrets management.
- 10. Maintain business continuity throughout migration.

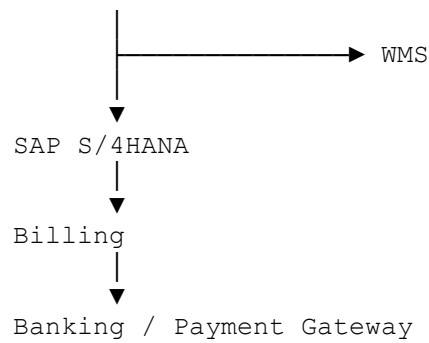
1.3 In-Scope Systems

System	Function	Integration Type	Migration Treatment
SAP S/4HANA	Core ERP	RFC / OData / SOAP	Hybrid
WMS	Warehouse Operations	REST / DB	Replatform
TMS	Transportation Management	REST / EDI	Refactor
Fleet Platform	Vehicle Tracking	REST / MQTT	Integrate
Customer Portal	Shipment Tracking	REST	Replatform
EDI Gateway	Trading Partners	AS2 / SFTP	Migrate
Billing Engine	Invoicing	REST / SQL	Refactor
BI Platform	Management Reporting	SQL / ETL	Rebuild
Document Management	POD / Customs Docs	SFTP / API	Migrate
Identity Platform	Enterprise IAM	SAML / OIDC	Integrate

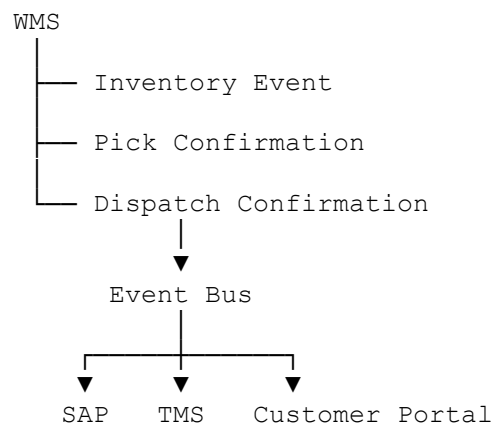
1.4 Business-Critical Integration Flows

Order-to-Cash





Warehouse Shipment Flow



2. Team Composition & Resourcing

NileTech deployed a cross-functional cloud transformation team combining **ERP specialists, Azure architects, integration engineers, DevSecOps, data engineers and logistics-domain consultants.**

2.1 Core Delivery Team

Role	Years of Exp	Certifications	Allocation %
Programme Director	18	PMP, PRINCE2 Practitioner	20%
Cloud Transformation Director	17	TOGAF, Azure Solutions Architect Expert	50%
Solution Architect	15	TOGAF, Azure Solutions Architect Expert	80%
Enterprise Integration Architect	14	Azure Integration Services, MuleSoft	100%
SAP Technical Architect	16	SAP Certified Technology Professional	70%
SAP Functional Consultant ×3	8–13	SAP S/4HANA Certifications	100%

Role	Years of Exp	Certifications	Allocation %
Azure Infrastructure Lead	12	Azure Administrator, Azure Solutions Architect	100%
Azure Engineers ×3	5–9	AZ-104, AZ-305	100%
Integration Engineers ×4	5–10	Azure Integration Services	100%
Data Migration Lead	11	Azure Data Engineer Associate	80%
Data Engineers ×2	6–8	DP-203	100%
DevSecOps Lead	11	AZ-400, CKA	100%
DevOps Engineers ×2	6–9	AZ-400, Terraform Associate	100%
Cybersecurity Architect	14	CISSP, CCSP	40%
Network Architect	15	CCNP Enterprise, Azure Network Engineer	50%
QA Lead	11	ISTQB Advanced	100%
QA Automation Engineers ×3	5–8	ISTQB Foundation	100%
Performance Engineer	9	ISTQB Performance Testing	50%
Data Governance Lead	12	CDMP	40%
Business Analysts ×3	7–12	CBAP	100%
Change Management Lead	13	Prosci, PMP	60%

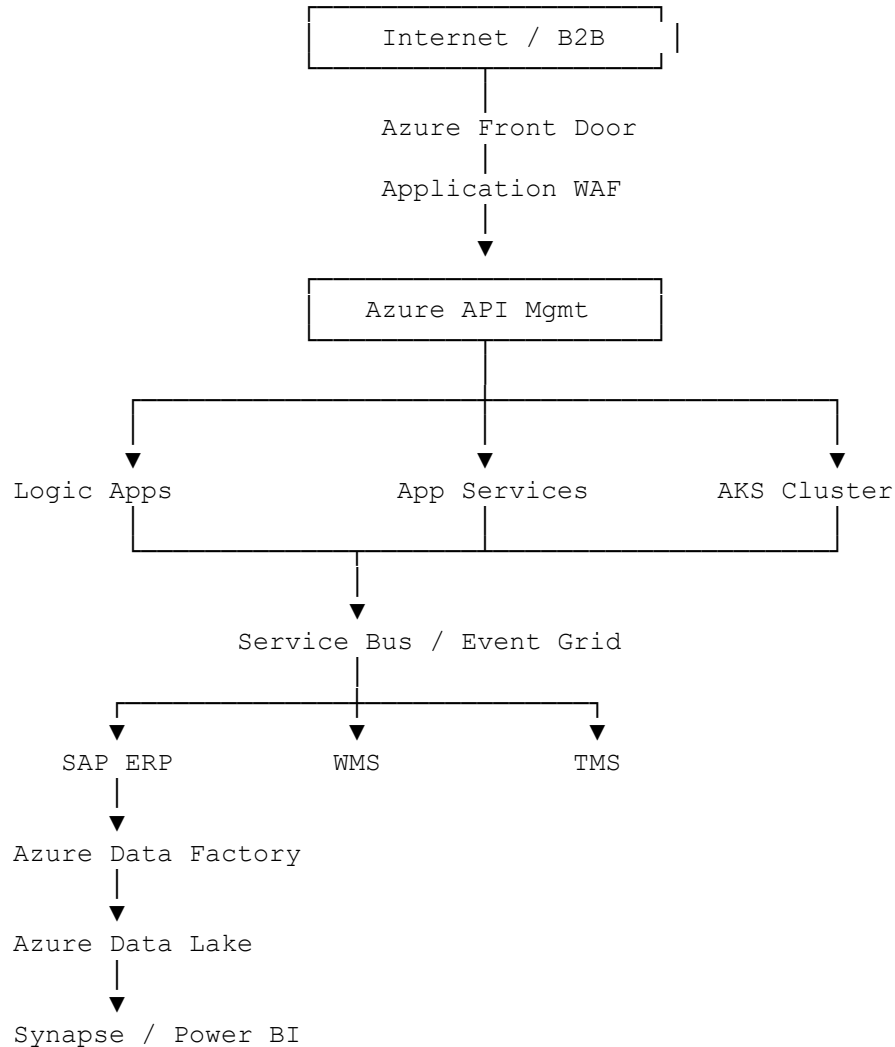
2.2 Resource Allocation by Workstream

Workstream	Peak FTE	Primary Roles	Allocation Window
Programme Management	2	PM / Programme Director	Weeks 1–44
Architecture	4	Solution / Cloud / Integration Architects	Weeks 1–30
ERP	5	SAP Consultants	Weeks 4–35
Azure Platform	6	Azure / DevOps Engineers	Weeks 5–38
Integration	5	Integration Engineers	Weeks 8–36
Data Migration	3	Data Engineers / Data Lead	Weeks 14–34
QA	5	QA / Automation / Performance	Weeks 20–39
Security	3	Cybersecurity / IAM	Weeks 5–40
Change & Training	3	Change Manager / SMEs	Weeks 30–44

3. Technical Architecture & Tech Stack

3.1 Target Azure Architecture

The target architecture was designed around **Azure Landing Zones, hub-and-spoke networking, private connectivity and managed integration services.**



3.2 Network Architecture

The Azure environment used a **hub-and-spoke topology**.

Hub

Responsible for shared enterprise services:

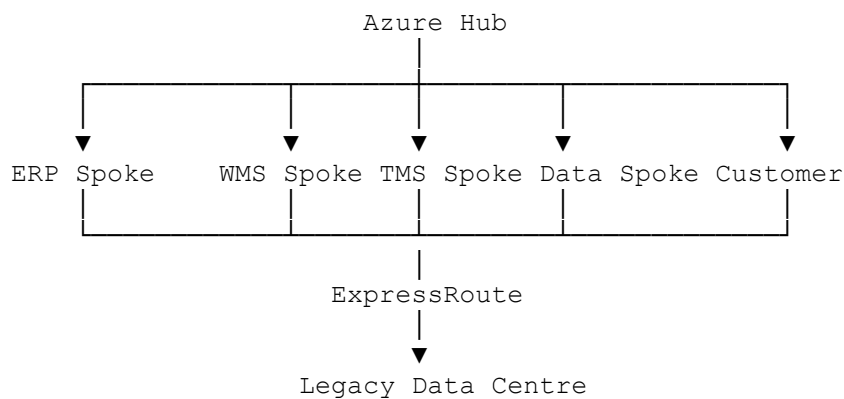
- Azure Firewall
- VPN / ExpressRoute termination
- Bastion
- DNS
- Azure Monitor

- Centralized security services

Spokes

Dedicated application boundaries:

- ERP Integration
- WMS
- TMS
- Customer Applications
- Data Platform
- Management / Reporting



3.3 Technology Stack

Architecture Layer	Technology	Primary Purpose
Cloud Platform	Microsoft Azure	Primary hosting platform
Landing Zone	Azure Landing Zones	Governance and subscription structure
Compute	Azure Kubernetes Service (AKS)	Containerized workloads
Serverless	Azure Functions	Event-driven processing
Integration	Azure Logic Apps	Workflow orchestration
API Management	Azure API Management	API exposure and governance
Messaging	Azure Service Bus	Reliable asynchronous messaging
Eventing	Azure Event Grid	Event distribution
ERP	SAP S/4HANA	Enterprise resource planning
ETL	Azure Data Factory	Data movement and migration
Data Lake	Azure Data Lake Storage Gen2	Central data repository
Analytics	Azure Synapse	Analytical workloads
BI	Power BI	Reporting and dashboards

Architecture Layer	Technology	Primary Purpose
Secrets	Azure Key Vault	Secret/certificate management
Identity	Microsoft Entra ID	Identity and access management
Security	Microsoft Defender for Cloud	Cloud security posture
SIEM	Microsoft Sentinel	Security monitoring
Monitoring	Azure Monitor / Application Insights	Observability
IaC	Terraform	Infrastructure provisioning
CI/CD	Azure DevOps	Build and release automation
Containers	Azure Container Registry	Container image management
Backup	Azure Backup	Data/workload protection
DR	Azure Site Recovery	Disaster recovery

3.4 Integration Standards

All newly developed integrations were required to comply with the following standards:

Synchronous Integration

- RESTful APIs.
- JSON payloads.
- OAuth 2.0 authentication.
- OpenAPI 3.x specifications.
- Standardized HTTP status codes.
- Correlation ID propagation.
- Idempotency keys for financial/operational transactions.

Asynchronous Integration

- Azure Service Bus.
- Event-driven processing.
- Dead-letter queues.
- Retry policies.
- Exponential backoff.
- Message deduplication.
- Poison-message handling.

File-Based Integration

Legacy B2B partners requiring file exchange were supported through:

- SFTP.
- AS2.

- PGP encryption.
- SHA-256 integrity validation.
- Automated file acknowledgement.
- Processing audit trails.

3.5 Migration Strategy

Each workload was classified according to a **6R migration model**.

Workload Category	Strategy	Example
Retire	Remove	Obsolete reporting server
Retain	Remain temporarily	Legacy SAP interfaces
Rehost	Lift & Shift	Existing application VM
Replatform	Managed Azure Service	SQL workload → Azure SQL
Refactor	Cloud-native redesign	Integration middleware
Replace	SaaS / Managed Service	Legacy monitoring

No production workload was migrated without completing:

1. Dependency discovery.
2. Data classification.
3. Security assessment.
4. Performance baseline.
5. Migration rehearsal.
6. Rollback procedure.
7. Business owner approval.

4. Implementation Timeline

The programme was structured into **44 weeks**, with infrastructure and integration workstreams progressing in parallel after architecture approval.

Phase	Deliverables	Duration in Weeks	Dependencies
1. Mobilisation	Governance, PMO, RAID register, stakeholder matrix	2	Contract award
2. Current-State Assessment	Application inventory, dependency map, integration catalogue	5	SME availability

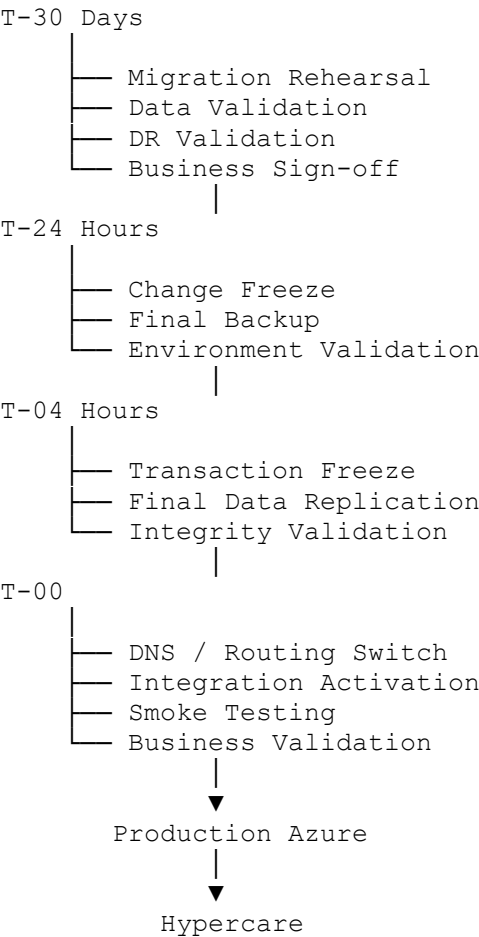
Phase	Deliverables	Duration in Weeks	Dependencies
3. Azure Landing Zone	Management groups, subscriptions, RBAC, policies, networking	6	Architecture approval
4. Security & Compliance Design	IAM model, security controls, data classification, threat model	5	Current-state assessment
5. Integration Architecture	API standards, event model, canonical data model	6	System inventory
6. Connectivity	ExpressRoute/VPN, DNS, firewall rules, private endpoints	5	Azure Landing Zone
7. ERP Integration Foundation	SAP connectivity, API layer, integration runtime	6	Connectivity
8. WMS/TMS Integration	Operational APIs, events, reconciliation processes	8	Integration foundation
9. Data Migration Factory	ETL pipelines, validation framework, migration scripts	8	Data model approval
10. Application Migration Wave 1	Low-risk workloads migrated	4	Landing Zone
11. Application Migration Wave 2	Business-critical workloads migrated	6	Wave 1 acceptance
12. Analytics Migration	Data Lake, Synapse, Power BI workloads	6	Data migration
13. SIT	End-to-end ERP/integration testing	5	Development completion
14. Performance & Resilience	Load, failover, DR and recovery testing	4	SIT completion
15. Security Validation	Penetration testing, vulnerability assessment, configuration review	4	Stable environment
16. UAT	Business acceptance, process validation	4	SIT + Security
17. Production Cutover	Migration execution, DNS switch, validation	2	UAT sign-off
18. Hypercare & Handover	Monitoring, stabilization, KT, operational acceptance	4	Production cutover

4.1 Migration Wave Structure

Wave	Workload Profile	Estimated Systems	Rollback Window
Wave 0	Non-production / Development	8	N/A
Wave 1	Low-criticality applications	12	24 hours

Wave	Workload Profile	Estimated Systems	Rollback Window
Wave 2	Medium-criticality applications	15	12 hours
Wave 3	High-criticality ERP integrations	10	8 hours
Wave 4	Customer-facing workloads	7	4 hours

4.2 Production Cutover Sequence



5. SLA & Support Matrix

The managed service agreement introduced a severity-based incident model covering the Azure platform, integration services and migrated applications.

Severity	Business Impact	Initial Response	Technical Engagement	Target Workaround	Target Restoration
Severity 1 — Critical	Complete outage affecting core logistics operations or ERP integration	15 min	≤30 min	2 hours	4 hours

Severity	Business Impact	Initial Response	Technical Engagement	Target Workaround	Target Restoration
Severity 2 — High	Major operational degradation affecting multiple sites/business units	30 min	≤1 hour	4 hours	8 hours
Severity 3 — Medium	Limited functionality degradation with operational workaround	2 hours	≤4 hours	1 Business Day	3 Business Days
Severity 4 — Low	Minor defect, reporting issue or service request	4 Business Hours	≤1 Business Day	N/A	5 Business Days / Planned Release

5.1 Azure Platform Support Matrix

Service Domain	Monitoring	Support Coverage	Primary Escalation
Azure Network	Azure Monitor + NSG Flow Logs	24×7	Cloud Engineering
API Management	Application Insights	24×7	Integration L2
Service Bus	Azure Monitor	24×7	Integration L2/L3
AKS	Azure Monitor / Container Insights	24×7	Platform Engineering
SAP Integration	Custom telemetry + SAP monitoring	24×7	ERP Integration Team
Data Factory	Pipeline Monitoring	24×7	Data Engineering
Data Lake	Azure Monitor	Business Hours + Critical 24×7	Data Platform
Security	Microsoft Sentinel	24×7	SOC
Identity	Entra ID Monitoring	24×7	IAM Team
Backup / DR	Azure Backup / ASR	24×7	Cloud Infrastructure

6. Compliance, Governance & Technical Controls

6.1 Azure Governance

All Azure resources were required to comply with the enterprise cloud governance baseline.

Mandatory controls included:

- **Azure Policy enforcement** at subscription/management-group level.
- Resource tagging using:

Environment
Application
BusinessUnit
CostCentre
DataClassification
Owner
Criticality

- No public IP exposure unless explicitly approved.
 - Private endpoints required for sensitive PaaS services.
 - Centralized logging enabled for production resources.
 - RBAC implemented using least-privilege principles.
 - Privileged operations required MFA.
 - Production administrative access required controlled privileged identities.
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6.2 Data Migration Controls

Each migration batch followed:

```
Extract
↓
Transform
↓
Validate
↓
Reconcile
↓
Load
↓
Post-Load Validation
↓
Business Sign-off
```

Validation included:

- Record-count reconciliation.
- Hash-based validation where applicable.
- Referential-integrity checks.
- Financial-total reconciliation.
- Duplicate detection.
- Mandatory-field validation.
- Business-rule validation.

Migration Acceptance Threshold

For critical ERP datasets:

100% reconciliation of financial balances and transactional control totals was mandatory before business acceptance.

Non-critical datasets were subject to agreed tolerance thresholds documented in the migration runbook.

7. Disaster Recovery Requirements

The target architecture incorporated regional recovery capabilities for business-critical services.

Workload	RTO	RPO	Recovery Mechanism
API Gateway	30 min	15 min	Azure-native configuration recovery
Integration Services	30 min	15 min	Geo-redundant deployment
Critical Applications	60 min	15 min	Azure Site Recovery
Operational Database	60 min	5–15 min	Geo-replication
Data Lake	4 hours	1 hour	GRS / backup
Reporting	8 hours	4 hours	Rebuild / replicated datasets

A DR exercise was required before final programme acceptance.

8. Final Acceptance Criteria

Production acceptance required completion of all mandatory gates:

Gate	Mandatory Condition
Architecture Gate	Approved HLD, LLD and integration architecture
Security Gate	No unresolved Critical security findings
Migration Gate	Reconciliation completed for all critical datasets
Performance Gate	Agreed transaction and API performance targets achieved
DR Gate	Recovery procedures successfully tested
UAT Gate	Business process owners approve migrated workflows
Operational Gate	Monitoring, alerting and runbooks operational
Handover Gate	Support team trained and knowledge transfer completed

Programme Outcome

The historical programme transitioned the logistics operator from a predominantly data-centre-centric integration model to an **Azure-based hybrid enterprise architecture**, with standardized APIs, asynchronous integration, centralized observability, automated infrastructure provisioning and controlled migration waves.

Historical Proposal Status: Awarded / Production Migration Completed / Managed Support Transitioned.